Zakary Raymond

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Technical Skills

Molecular Biology: qPCR, RT-qPCR, Western Blotting, SDS-PAGE, DNA/RNA Extraction, Primer Design, BLAST, Gel Electrophoresis, Immunostaining

Genomics & Cloning: Vector Cloning, Plasmid Assembly, Restriction Enzyme Digestion, Transformation, Colony Screening

Cell Culture & Tissue Models: Mammalian Cell Culture, Tissue Culture, Primary Corneal Epithelial Cells (rabbit harvest & dissection), Adipose-Derived Stem Cells (ADSCs), iPSCs, ADSC Differentiation (Lean/Obese/Lipedema), Endothelial Cells (HUVEC/flow chamber), Aseptic Technique, Passaging, Cryopreservation

Analytical & Assay: HPLC Purification, MIC Assays, Chemical Fractionation, Liquid-Liquid Partition, Antimicrobial Biofilm Inhibition Screening, Dose-Response Analysis, Inoculation & Plate Streaking

Imaging & Analysis: Confocal Microscopy, High-Content Imaging (HCI), CellProfiler Pipelines, FIJI/ImageJ, SEM + Sputter Coating, Quantitative Biomarker Activation Analysis

Software & Data: Benchling (GxP-aligned ELN), Python (Image Analysis, Automation), MATLAB, RStudio, GraphPad Prism, LabVIEW, COMSOL Multiphysics

Fabrication & Devices: Electrospinning, Scaffold Fabrication, Microfluidics, SolidWorks CAD, 3D Printing (FDM/Bioprinting), Resin Printing, CNC Machining, Laser Cutting, Marlin Firmware

Regulatory & Ops: GxP Workflows, ELN Documentation, SOPs, Sterility & Contamination Control, FDA Regulatory Awareness (510(k)/PMA pathway exposure)

Research Experience

Vascular Mechanobiology Researcher — May 2025 - Jul 2025
Worcester Polytechnic Institute (NSF REU) — *Worcester, MA*

- Increased **High-Content Imaging (HCI)** throughput **40%** across **50+ batch-processed images** by building a custom **Python** and **FIJI/ImageJ** image-analysis pipeline
- Characterized endothelial response across **3 experimental conditions** via **6-hour flow chamber assays** with **immunostaining**, **confocal microscopy**, and **CellProfiler/GraphPad Prism** quantification
- Maintained **100% sterility** across all summer trials by culturing mammalian endothelial cells under **aseptic technique** and contamination-control protocols

Corneal Tissue Researcher — Aug 2024 - May 2026
University of the Pacific — *Stockton, CA*

- Established a validated **Western blotting** protocol for corneal tissue models by optimizing **SDS-PAGE** and immunoblotting conditions
- Sustained **zero contamination events** across **5-month** corneal epithelial cell cultures by maintaining mammalian tissue culture through rabbit corneal harvest, dissection, and aseptic passaging
- Targeted a **10 μm fiber diameter** by fabricating **electrospun** collagen/gelatin/PLLA scaffolds and characterizing fiber morphology via **SEM**

Analytical Chemistry Researcher — Jan 2025 - May 2026
University of the Pacific — *Stockton, CA*

- Achieved **100% analyte purity** by performing **HPLC purification** across all isolated fractions
- Screened **150+ drug candidates** for antimicrobial biofilm inhibition by running **MIC assays**, dose-response analysis, chemical fractionation, and liquid-liquid partition
- Maintained full sample traceability across **200+ samples** by executing inoculation, plate streaking, and **Benchling-based GxP-aligned ELN** documentation

Adipose Stem Cell Researcher — Feb 2024 - Aug 2024
University of the Pacific — *Stockton, CA*

- Verified **ADSC differentiation** across **3 donor lineages** (lean/obese/lipedema) and **2 induction variations** by validating outcomes via **qPCR**, **RT-qPCR**, and **Western blotting** analyzed in **RStudio**
- Communicated findings to **2 national conference audiences** by authoring a 10-page technical summary on ADSC differentiation outcomes

Projects

Printess Bioprinter | *Python, Marlin Firmware, SolidWorks* — Nov 2025 - present

- Achieved **±5 μm motion repeatability** on a custom dual-syringe extrusion bioprinter using a **RUMBA+ board** and **Marlin firmware**

Wheelchair-Gurney Senior Project | *SolidWorks, CNC Routing, Plasma Cutting* — Aug - Dec 2025

- Designed a **4-bar linkage** transfer mechanism targeting **~5° ROM** and developed a **B2B/D2C go-to-market plan**

Genetics & Molecular Cloning Lab | *BLAST, Primer Design, Restriction Enzyme Digestion, Transformation* Aug - Dec 2025

- Assembled **2 plasmid constructs** and confirmed **≥1 ng/µL DNA/RNA** purity via **BLAST primer design** and colony screening

Wind Vehicle | *UTS* Aug - Dec 2023

- Built and tested a wind-powered vehicle to **ISO 1100 standards** across **2 configurations**

PRESENTATIONS

TNF-α Disrupts Cooperative vWF Anchoring by P-selectin & Endothelial Glycocalyx HS | WPI Poster Symposium 2025; Stanford Best Poster (Bay Area Cardiovascular Symposium); BMES 2025 (San Diego); Harvard NCRC 2026
Optimization of Adipogenic Differentiation of ADSCs | Harvard NCRC 2025; NCUR 2026 (Pittsburgh)
Screening Natural Product Extracts for Bacterial Biofilm Inhibition Activity | UOP Research & Creativity Showcase 2026 (200+ audience)
MIT Grand Hacks 2026, Trauma & Rehabilitation Track | Wearable gas sensor concept; FDA pathway + B2C commercialization model

EDUCATION

University of the Pacific Stockton, CA
B.S. Biomedical Engineering, Minor: Biology | **GPA: 3.67 overall / 3.87 major**
Honors: Tau Beta Pi, NSF REU, Pacific SURF 2024, Stanford Best Poster (Bay Area Cardiovascular Symposium), Yale PATHS, Segal Award, BMES 2025, NCUR 2025, NCRC 2025 & 2026, MIT Grand Hacks 2026
University of Technology Sydney Sydney, AU
Exchange Student - Mechanical & Biomedical Engineering Fall 2023

LEADERSHIP & SERVICE

University of the Pacific Stockton, CA
VP, Tau Beta Pi Jan 2025 - May 2026

- Inducted **30+ members** and grew chapter programming **200%** by serving on the NEST committee and expanding events

President & VP Recruitment, Beta Theta Pi May 2024 - Dec 2025

- Led a 30-member chapter to **top national recognition** among **140+ chapters** with **95% retention**

Professional Development Chair, Theta Tau Dec 2025 - May 2026

- Ran **4+ industry events** for **30+ members**

Event Coordinator, BMES May 2025 - May 2026

- Secured **$6,000 in hackathon funding** for a dog prosthetic project

Makerspace Steward, UOP Feb 2024 - May 2025

- Supported **50+ students** across **100+ hours** and reduced equipment downtime **10%**

Residential Assistant, UOP Aug 2024 - May 2026

- Supported **120+ students** on a **$4,000** programming budget; earned Mentor Tiger and Big Tiger awards

College Corps Fellow, UOP Aug 2022 - Jul 2023

- Delivered **450+ service hours** working with **200+ K-8 students**